

SAFETY DATA SHEET

KodaPure TNA Extraction Kit

For microbe, animal tissue, and soil TNA extraction

RESEARCH USE ONLY

Not for diagnostic, therapeutic, food, household, or human/animal consumption use.

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01 Identification

Field	Information
Product identifier	KodaPure TNA Extraction Kit
Product type	Laboratory reagent kit containing aqueous buffers, proprietary lysis and wash buffers, alcohol-containing buffers, magnetic beads, and homogenizing beads.
Recommended use	TNA extraction from microbes, animal tissue, and soil samples.
Restrictions on use	Research use only. Not for diagnostic, therapeutic, food, household, human consumption, or animal consumption use.

02 Hazard Identification

This kit contains separate bottles and hazards differ by buffer. Handle each bottle according to its label and the table below.

Buffer	Signal word	Pictogram	Main hazard	Key warning
PW	WARNING		Mild alkaline irritant due to alkaline buffer and surfactant.	Avoid skin and eye contact.
LBI	WARNING		Contains a proprietary chaotropic salt. Harmful if swallowed. Causes skin and eye irritation.	Do not mix with bleach or hypochlorite.
LBII	WARNING		Contains a detergent/surfactant blend. Causes skin and eye irritation.	Avoid contact and aerosols.
LBIII	WARNING		Contains proprietary plant lysis detergent/surfactant and salts. Causes skin and eye irritation.	Avoid contact and aerosols. Mix before use.
Tube A	NOT CLASSIFIED	—	Dry inhibitor-removal/adsorbent reagent. Low hazard under normal laboratory use; dust may irritate if dispersed.	Avoid inhaling dust.

Buffer	Signal word	Pictogram	Main hazard	Key warning
MB	NOT CLASSIFIED	—	Magnetic bead suspension. Low hazard under normal laboratory use.	Avoid ingestion and inhalation.
BB	DANGER	 	Contains flammable alcohol. Highly flammable liquid and vapor. Causes serious eye irritation. May cause drowsiness or dizziness.	Keep away from ignition sources.
WBII	DANGER	 	Contains flammable alcohol. Highly flammable liquid and vapor. Causes eye irritation.	Keep away from ignition sources.
WBIII	WARNING	 	Contains proprietary chaotropic salt and flammable alcohol. Harmful/irritant and flammable.	Do not mix with bleach. Keep away from ignition sources.
EB	NOT CLASSIFIED	—	Aqueous elution buffer. Low hazard under normal use.	Avoid ingestion.

- ▶ Do not mix chaotropic salt-containing buffers or waste with bleach or hypochlorite solutions. Toxic gases may form.
- ▶ BB, WBI, WBII, and WBIII contain alcohol and should be kept away from heat, sparks, open flames, and hot surfaces.
- ▶ Treat all biological samples and used consumables as potentially infectious.

03 Composition / Information on Ingredients

This section lists hazardous ingredient classes needed for safe handling. Exact chemical identities and concentrations are withheld as proprietary formulations where applicable.

Buffer	Ingredient	Range / disclosure	Hazard note
PW	Proprietary alkaline salt	Proprietary formulation	Alkaline irritant
PW	Proprietary phosphate salt	Proprietary formulation	Irritant
PW	Inorganic salt	Proprietary formulation	Not classified at supplied concentration
PW	Adsorbent polymer	Proprietary formulation	Dust may irritate if dried
PW	Non-ionic surfactant	<1%	May irritate at higher concentration
LBI	Proprietary chaotropic salt	30–50%	Harmful, irritant
LBI	Chelating agent	Proprietary formulation	Low hazard at supplied concentration
LBI	Buffering agent	Proprietary formulation	Low hazard at supplied concentration
LBII	Anionic detergent	1–5%	Skin and eye irritant

Buffer	Ingredient	Range / disclosure	Hazard note
LBII	Non-ionic surfactant	Proprietary formulation	Irritant. Aquatic hazard at higher concentration
LBII	Chelating agent	Proprietary formulation	Low hazard at supplied concentration
LBII	Buffering agent	Proprietary formulation	Low hazard at supplied concentration
LBIII	Proprietary detergent/surfactant blend	Proprietary formulation	Skin and eye irritant
LBIII	Proprietary salts	Proprietary formulation	May irritate at supplied concentration
LBIII	Chelating/buffering agents	Proprietary formulation	Low hazard at supplied concentration
Tube A	Adsorbent/inhibitor-removal material	Proprietary formulation	Dust may irritate if dispersed
MB	Magnetic beads	Suspension	Low hazard under normal use
BB	Flammable alcohol	>90%	Highly flammable, eye irritant, STOT SE 3
WBII	Flammable alcohol	70–90%	Highly flammable, eye irritant
WBIII	Proprietary chaotropic salt	10–30%	Harmful, irritant
WBIII	Flammable alcohol	40–60%	Flammable, eye irritant
WBIII	Buffering agent	Proprietary formulation	Low hazard at supplied concentration
EB	Chelating agent	Proprietary formulation	Low hazard at supplied concentration
EB	Buffering agent	Proprietary formulation	Low hazard at supplied concentration

04 First-Aid Measures

Exposure	First aid
Eye contact	Rinse carefully with water for at least 15 minutes. Remove contact lenses if easy to do. Continue rinsing. Seek medical advice if irritation persists.
Skin contact	Remove contaminated clothing. Wash skin with soap and plenty of water. Seek medical advice if irritation persists.
Inhalation	Move the affected person to a new place with fresh air, comfortable for breathing. Seek medical advice if symptoms occur.
Ingestion	Rinse mouth. Do not induce vomiting. Seek medical advice. Never give anything by mouth to an unconscious person.
Most important symptoms	Eye irritation, skin irritation, nausea, dizziness, drowsiness, headache, or gastrointestinal discomfort.

05 Fire-Fighting Measures

Item	Information
Suitable extinguishing media	Alcohol-resistant foam, dry chemical, carbon dioxide, or water spray.
Specific hazards	BB, WBII, and WBIII contain flammable alcohol. Vapors may form explosive mixtures with air. Combustion may produce carbon oxides, nitrogen oxides, acidic fumes, and irritating fumes.
Firefighter protection	Wear self-contained breathing apparatus and full protective gear. Cool exposed containers with water spray.
Unsuitable media	Do not use a direct high-pressure water jet on alcohol fires.

06 Accidental Release Measures

Item	Action
Personal precautions	Wear gloves, lab coat, and eye protection. Avoid breathing vapors or aerosols. Remove ignition sources for alcohol-containing buffers.
Environmental precautions	Prevent release to drains where possible.
Spill cleanup	Absorb with inert material or paper towel. Clean area with water and detergent. Do not use bleach on LBI or WBIII spills.

07 Handling and Storage

- Use only in a laboratory area with good ventilation.
- Wear nitrile gloves, lab coat, and eye protection.
- Keep BB, WBII, and WBIII away from heat, sparks, open flames, and hot surfaces.
- Do not mix LBI, WBIII, or related waste with bleach or hypochlorite solutions.
- Keep bottles tightly closed when not in use.
- Store all buffers and beads at room temperature unless the product label states otherwise. MB may be stored longer at 4 °C.
- Treat biological samples and used consumables as potentially infectious.

08 Exposure Controls / Personal Protection

Control	Requirement
Engineering controls	Use in a well-ventilated laboratory. Use a fume hood if handling larger volumes or if alcohol vapors are noticeable.
Eye / face protection	Safety glasses or goggles.
Skin protection	Nitrile gloves and lab coat. Change gloves if contaminated.
Respiratory protection	Not normally required for small-volume kit use with adequate ventilation.
Hygiene measures	Do not eat, drink, or apply cosmetics in the work area. Wash hands after use.

09 Physical and Chemical Properties

Property	Value
Physical state	Liquids, bead suspensions, and dry homogenizing beads.
Color	Mostly clear/colorless. Magnetic bead suspension appears dark/black.
Odor	Alcohol odor for BB, WBII, and WBIII.
pH	PW approximately pH 10. LBI, LBII and LBIII approximately pH 8. WBIII pH 7.5. EB approximately pH 8.
Flash point	Alcohol-containing buffers are flammable. Exact flash point not determined for mixtures.
Solubility	Buffers are water-miscible. Magnetic beads and homogenizing beads are insoluble particles.
Viscosity	Not determined.
Density	Not determined.

10 Stability and Reactivity

Item	Information
Reactivity	No dangerous reaction expected under normal laboratory handling when used as directed.
Chemical stability	Stable under recommended storage conditions.
Possibility of hazardous reactions	Chaotropic salt-containing buffers may react with bleach/hypochlorite to release toxic gases. Alcohol vapors may ignite.
Conditions to avoid	Heat, sparks, open flames, hot surfaces, evaporation to dryness near ignition sources, and incompatible disinfectants.
Incompatible materials	Bleach, hypochlorite solutions, strong oxidizers, strong acids, and strong bases.
Hazardous decomposition products	Carbon oxides, nitrogen oxides, acidic fumes, and other irritating or toxic fumes under fire or improper mixing conditions.

11 Toxicological Information

Endpoint	Information
Likely exposure routes	Skin, eyes, inhalation of vapors/aerosols, accidental ingestion.
Acute toxicity	LBI and WBIII contain proprietary chaotropic salts and may be harmful if swallowed. Alcohol-containing buffers may cause central nervous system effects at high vapor exposure.
Skin corrosion / irritation	PW, LBI, LBII, LBIII, and WBIII may irritate skin. Detergent-containing buffers are irritating.
Serious eye damage / irritation	Alcohol-containing buffers, chaotropic salt-containing buffers, and detergent-containing buffers may irritate eyes.

Endpoint	Information
Respiratory irritation	Alcohol vapors may cause dizziness or drowsiness. Aerosols may irritate the respiratory tract.
Sensitization	No data available for formulated kit mixtures.
Carcinogenicity	No component is expected to be classified as carcinogenic based on available formulation information and common laboratory use.

12 Ecological Information

- Do not discharge buffers directly to the environment or drains unless permitted by local rules.
- Detergents and surfactants may be harmful to aquatic life at sufficient concentration.
- Alcohols are expected to be biodegradable but may present oxygen demand and flammability hazards in bulk.
- No full ecological testing has been performed on the final kit mixtures.

13 Disposal Considerations

- Dispose of unused buffers, used tips, tubes, and sample waste according to local chemical and biological waste rules.
- Do not add bleach to LBI, WBIII, or waste containing these buffers.
- Alcohol-containing waste should be treated as flammable chemical waste where applicable.
- Biological samples should be inactivated or disposed of as biohazard waste according to institutional procedures.

14 Transport Information

Transport item	Information
UN number	UN3316 may apply when shipped as a regulated chemical kit. Classification must follow final packaging and local transport rules.
Proper shipping name	Chemical kit, where applicable. Alcohol-containing buffers may trigger flammable liquid or chemical kit transport entries.
Transport hazard class	Class 9 may apply under UN3316, depending on final packaging and applicable transport rules.
Packing group	II or III may apply under UN3316, depending on final packaging and applicable transport rules.
Environmental hazards	Not classified as marine pollutant based on available formulation information.
Special precautions	Keep away from heat and ignition sources. Protect bottles from breakage and leakage.

15 Regulatory Information

- Prepared in 16-section SDS format for laboratory chemical products.
- Regulatory classification may vary by country, concentration cutoffs, final packaging volume, and shipping configuration.
- KreatBio Sdn. Bhd., as manufacturer and supplier, is responsible for product hazard communication. Users are responsible for local chemical storage, waste, and workplace safety compliance.
- For research use only.

16 Other Information

- This SDS was prepared by KreatBio Sdn. Bhd. based on available formulation and hazard information as of the revision date.
- Exact formulation details are confidential. Hazard classifications and safe-handling instructions are provided for customer use.
- Information applies only to the product as supplied and used according to the product protocol.

End of Safety Data Sheet KreatBio · KodaPure TNA Extraction Kit · Confidential